

Pranav Karthik

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EDUCATION

University of Wisconsin-Madison

Graduating May 2027

B.S. in Computer Science and Data Science | **GPA: 3.7**, Dean's List, Academic Coach for Math

Relevant Coursework: Data Structures/Algorithms (Competitive Programming), OOP, Advanced Java/Python/R, Linear Algebra, Artificial Intelligence

Affiliations: AI Club @ UW-Madison – Leading 8-person team shipping PyTorch realtime video editing pipeline

WORK EXPERIENCE

Sprocket Lab

Feb. 2026 – Present

AI/ML Researcher @ UW Madison

- Scraped **15K+** Polymarket events and created a benchmark suite for LLM performance on real-world forecasting.
- Fine-tuning an LLM with reinforcement learning in an **autonomous model loop** that generates search queries, retrieves and reasons over live web data, and refines predictions, achieving a **31%** Brier score reduction versus baseline models on 400+ held-out resolved markets.

Speech Processing and Auditory Neuroscience Lab

Aug. 2025 – Jan 2026

Research Assistant @ UW-Madison

- Trained neural networks to distinguish how sound tones are encoded in brain activity, with a **40%** increase in classification accuracy over traditional Fourier-transform baselines on EEG datasets with **10K+** time-series records.
- Built a **98%** accurate pupil-tracker on Raspberry Pi that monitors subject attention during EEG experiments.

Loyltz Rewardz

May 2025 – Aug. 2025

Full Stack Software Engineering Intern

- Built a **React/TypeScript** admin dashboard, centralizing workflows across **20+** merchant websites which reduced credit rewards deployment times from **2 hours to 5 minutes**.
- Deployed Spring Boot REST endpoints benchmarked to handle **6K+ requests/sec** and implemented JWT auth and JUnit tests for reliability under load.
- Built a CI/CD pipeline that improved development productivity by **35%** using Docker and Github Actions.
- Led the software development life cycle **end-to-end** in a 3-person Agile team: gathered requirements, designed APIs and UI, tested, and shipped to production with post-release iteration.

IndiaIdeas.Ltd, BillDesk

June 2023 – Aug. 2023

Software Engineering Intern — Real-Time Systems

- Owned backend development for high-volume payment processing services in **Spring Boot** and **SQL**, supporting **1.3M+** API requests per day across critical transaction workflows.
- Implemented secure authentication and secret-management workflows using **OAuth 2.0** token validation and reduced invalid token requests by **40%**.
- Maintained **99.9%** production availability by building internal monitoring dashboards using Prometheus/Grafana and deployment tooling via Docker.

PROJECTS

Haven – Disaster Relief Web App | TypeScript, Node.js, Supabase, AWS Lambda, Langchain

- Built an SMS-based disaster response app that allows users to text “HELP” to receive live details and directions to the nearest verified open shelter, cutting emergency lookup time from **5 minutes to 30 seconds**.
- Designed data pipelines to ingest and normalize **50K+** public shelter, weather, and road-closure events with TypeScript, AWS Lambda and Supabase, processing **4K+** live updates per day.
- Implemented a RAG system with Langchain, enabling natural-language queries with **98% retrieval accuracy**.
- Shipped to **400+** beta users a production-ready React/Node.js frontend with real-time shelter maps.

SHIPWizard – Student Health Insurance Assistant | Python, FastAPI, React, N8N

- Developed a React/FastAPI insurance claims copilot utilizing **60+ policy documents** to streamline how students understand health insurance benefits and file reimbursement.
- Engineered context-aware **N8N** workflows that use **NLP** to parse claim descriptions, retrieve the right policy sections using RAG, and auto-generate prefilled forms which cut manual data entry by **94%**.
- Fine-tuned a **PyTorch** model on claim context and policy data to intelligently understand nuances and autofill forms, resulting in **92%** of generated claims meeting insurance approval criteria on first submission.

SKILLS

Languages: Java, Python, SQL, TypeScript, JavaScript, R, C++

Technologies/Frameworks: Spring Boot, PyTorch, React, Node.js, Docker, Kubernetes, AWS, Apache Spark/Kafka, REST APIs